



Started on Friday, 25 September 2026, 3:57 PM

State Finished

Completed on Friday, 25 September 2026, 4:07 PM

Time taken 10 mins 43 secs

Marks 1.00/1.00

Grade 4.00 out of 4.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int n;
4      scanf("%d", &n);
5      int a[n];
6      for (int i = 0; i < n; i++) {
7          scanf("%d", &a[i]);
8      }
9      for (int i = 0; i < n; i++) {
10         for (int j = i + 1; j < n; j++) {
11             if (a[i] == a[j]) {
12                 printf("%d", a[i]);
13                 return 0;
14             }
15         }
16     }
17     return 0;
18 }
19
20

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Friday, 25 September 2026, 4:08 PM

State Finished

Completed on Friday, 25 September 2026, 4:10 PM

Time taken 2 mins 16 secs

Marks 1.00/1.00

Grade 4.00 out of 4.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1 | #include <stdio.h>
2 | int main() {
3 |     int n;
4 |     scanf("%d", &n);
5 |     int a[n];
6 |     for (int i = 0; i < n; i++) {
7 |         scanf("%d", &a[i]);
8 |     }
9 |     for (int i = 0; i < n; i++) {
10 |         for (int j = i + 1; j < n; j++) {
11 |             if (a[i] == a[j]) {
12 |                 printf("%d", a[i]);
13 |                 return 0;
14 |             }
15 |         }
16 |     }
17 |     return 0;
18 | }
19 |

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Friday, 25 September 2026, 4:10 PM

State Finished

Completed on Friday, 25 September 2026, 4:19 PM

Time taken 8 mins 45 secs

Marks 1.00/1.00

Grade 30.00 out of 30.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int T;
4      scanf("%d", &T);
5      while (T--) {
6          int n1, n2;
7          scanf("%d", &n1);
8          int a[n1];
9          for (int i = 0; i < n1; i++) {
10             scanf("%d", &a[i]);
11         }
12         scanf("%d", &n2);
13         int b[n2];
14         for (int i = 0; i < n2; i++) {
15             scanf("%d", &b[i]);
16         }
17         int i = 0, j = 0;
18         int first = 1;
19         while (i < n1 && j < n2) {
20             if (a[i] == b[j]) {
21                 if (first == 0) {
22                     printf(" ");

```

```

23         }
24         printf("%d", a[i]);
25         first = 0;
26         i++;
27         j++;
28     }
29     else if (a[i] < b[j]) {
30         i++;
31     }
32     else {
33         j++;
34     }
35 }
36 printf("\n");
37 }
38
39 return 0;
40 }
41

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Friday, 25 September 2026, 4:20 PM

State Finished

Completed on Friday, 25 September 2026, 4:23 PM

Time taken 2 mins 54 secs

Marks 1.00/1.00

Grade 30.00 out of 30.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int T;
5      scanf("%d", &T);
6
7      while (T--) {
8          int n1, n2;
9          scanf("%d", &n1);
10
11          int a[n1];
12          for (int i = 0; i < n1; i++)
13              scanf("%d", &a[i]);
14
15          scanf("%d", &n2);
16
17          int b[n2];
18          for (int i = 0; i < n2; i++)
19              scanf("%d", &b[i]);
20
21          int i = 0, j = 0;
22

```

```

23 ~
24 ~
25 ~
26 ~
27 ~
28 ~
29 ~
30 ~
31 ~
32 ~
33 ~
34 ~
35 ~
36 ~
37 ~
38 ~
39 ~
40 ~

```

```

while (i < n1 && j < n2) {
    if (a[i] == b[j]) {
        printf("%d ", a[i]);
        i++;
        j++;
    }
    else if (a[i] < b[j])
        i++;
    else
        j++;
}

printf("\n");
}

return 0;
}

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Friday, 25 September 2026, 4:24 PM

State Finished

Completed on Friday, 25 September 2026, 4:26 PM

Time taken 2 mins 32 secs

Marks 1.00/1.00

Grade 4.00 out of 4.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     int i = 0, j = 1;
15
16     while (i < n && j < n) {
17         if (a[j] - a[i] == k) {
18             printf("1");
19             return 0;
20         }
21
22         if (a[j] - a[i] < k)
23             j++;
24         else
25             i++;
26
27         if (i == j)
28             j++;
29     }
30
31     printf("0");
32
33     return 0;
34 }
35

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Friday, 25 September 2026, 4:26 PM

State Finished

Completed on Friday, 25 September 2026, 4:30 PM

Time taken 4 mins 1 sec

Marks 1.00/1.00

Grade 4.00 out of 4.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     for (int i = 0; i < n; i++) {
15         for (int j = i + 1; j < n; j++) {
16             if (a[j] - a[i] == k) {
17                 printf("1");
18                 return 0;
19             }
20         }
21     }
22
23     printf("0");
24
25     return 0;
26 }
27

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓

	Input	Expected	Got	
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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